

Customer Shopping Behavior Analysis

End-to-end data analytics project – from raw transactional data to actionable business recommendations using Python, SQL Server, and Power BI.



Dataset at a Glance

3,900

Transactions

Full purchase records analyzed

18

Columns

Customer, purchase & behavior data

37

Missing Ratings

Imputed using category median

Customer Data

Age, Gender, Location, Subscription Status

Purchase Data

Item, Category, Amount, Season, Size, Color

Behavior Data

Discount, Promo Code, Previous Purchases, Frequency, Review Rating, Shipping Type

Analytical Workflow



The project flows from raw data preparation in Python, through business-oriented SQL analysis, to an interactive Power BI dashboard — ensuring reproducibility and decision-ready outputs.

Data Preparation in Python

Exploration

Loaded via pandas; structure reviewed with `df.info()` and `.describe()`.

Missing Values

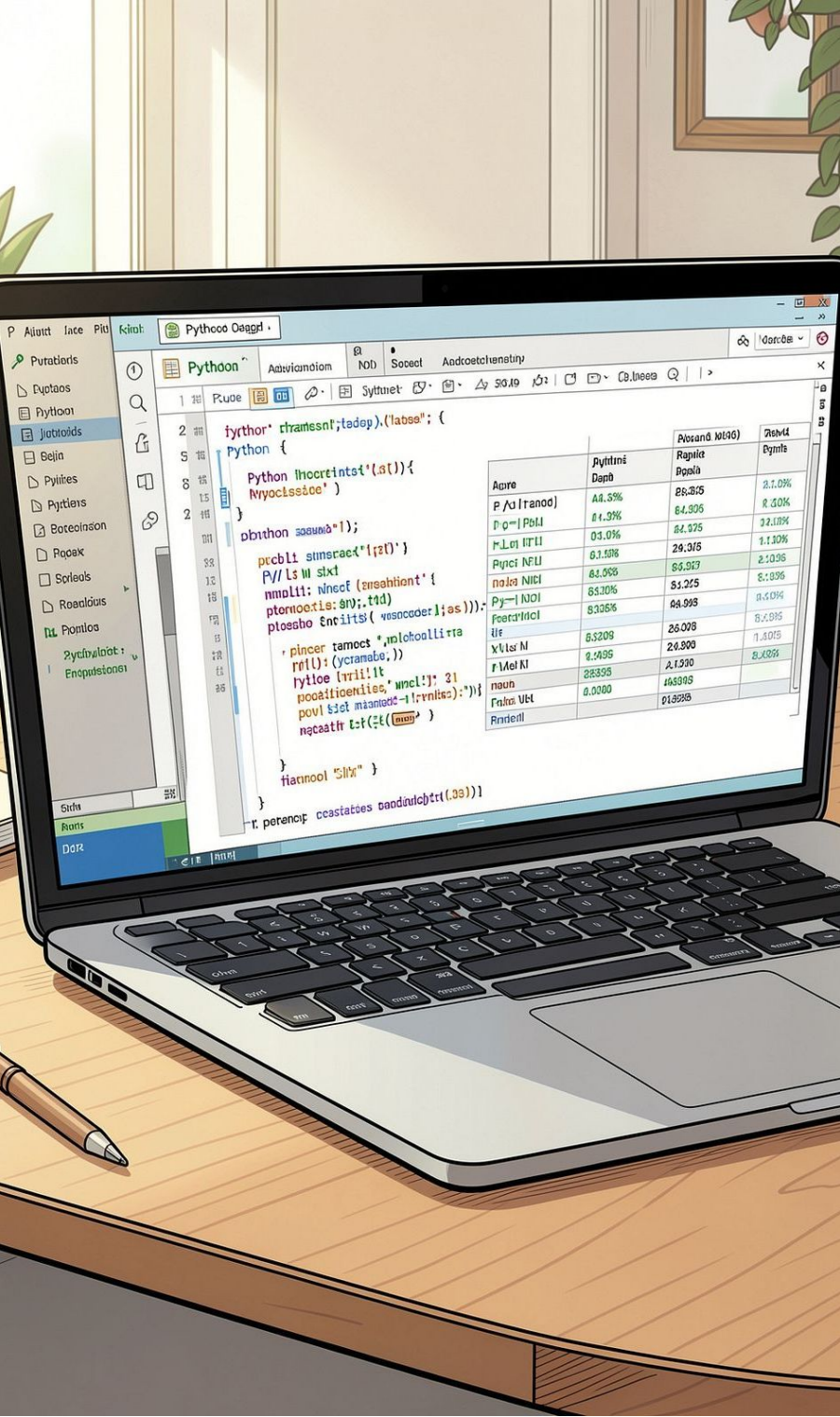
37 null Review Ratings imputed using category-level median.

Feature Engineering

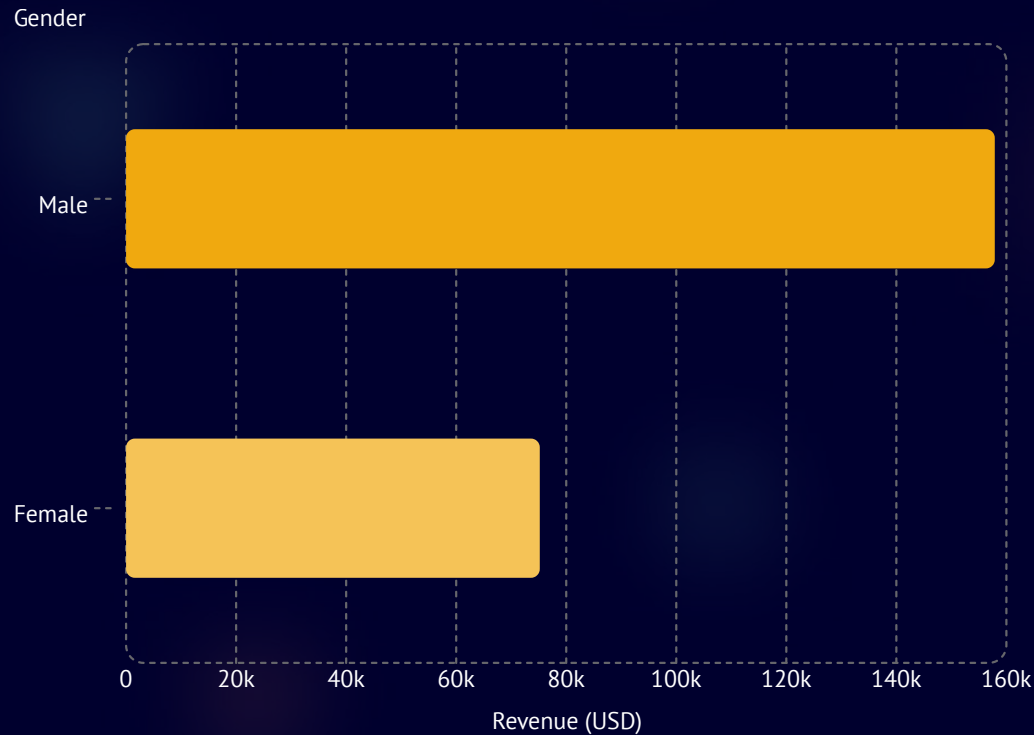
Created `age_group` (Young Adult → Senior) and `purchase_frequency_days` fields.

Consistency & Load

Removed redundant `promo_code_used` column; loaded cleaned DataFrame into SQL Server.



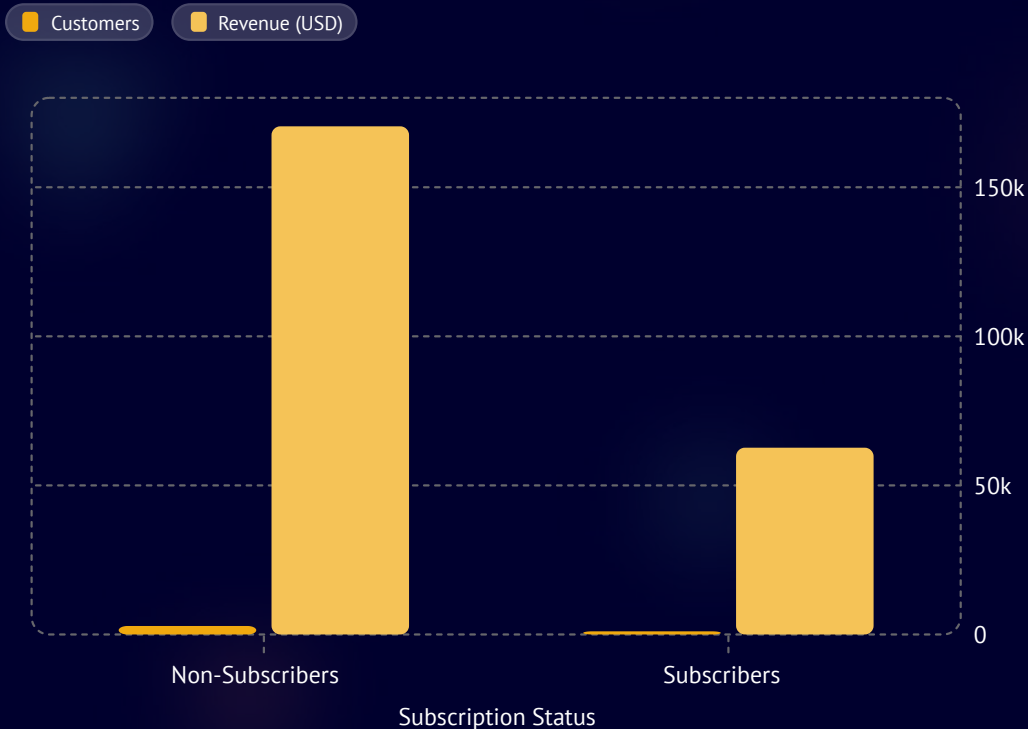
SQL Analysis — Revenue & Ratings



Key Findings

- **Male customers** generated **\$157,890** in revenue vs. \$75,191 for females.
- **Top-rated products:** Gloves (3.86), Sandals (3.84), Boots (3.82), Hat (3.80), Skirt (3.78).
- **Express shipping** averaged \$60.48 vs. \$58.46 for Standard.

Subscriptions & Customer Segments



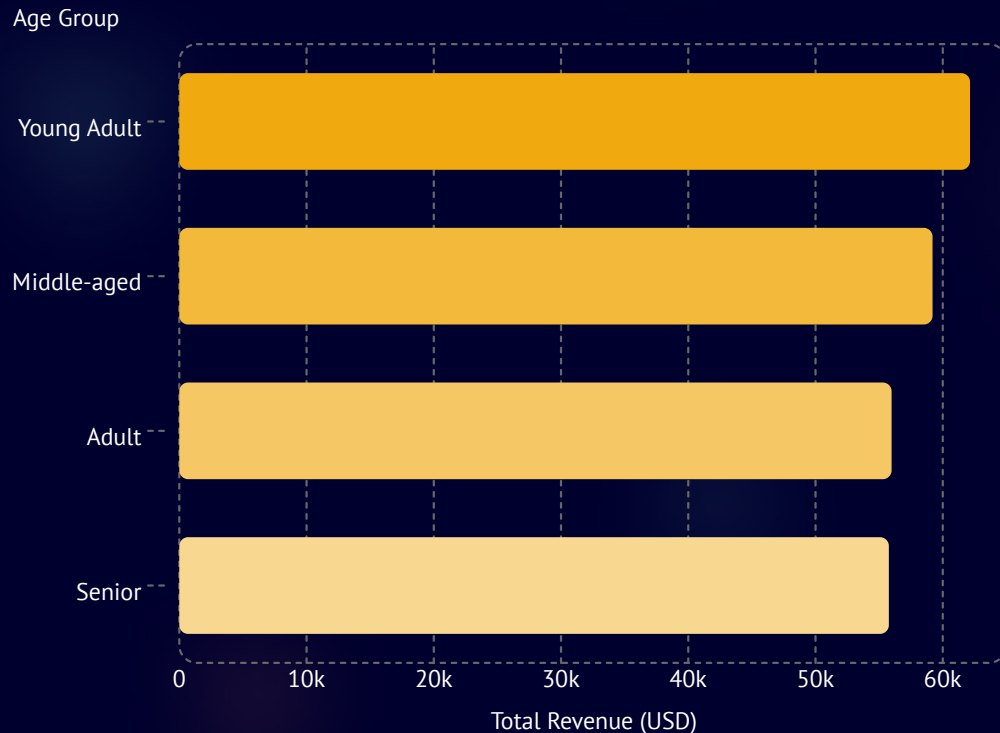
Segment Distribution

- **Loyal:** 3,116 customers (dominant segment)
- **Returning:** 701 customers
- **New:** 83 customers

Repeat Buyers

958 subscribers vs. 2,518 non-subscribers in the >5 purchase group. Discount dependence is high for Hat (50%), Sneakers (49.66%), and Coat (49.07%) – useful for targeted retention offers.

Product Rankings & Age Group Revenue



Top Products by Category

- **Accessories:** Jewelry (171), Belt (161), Sunglasses (161)
- **Clothing:** Blouse (171), Pants (171), Shirt (169)
- **Footwear:** Sandals (160), Shoes (150)

Young Adults lead revenue at \$62,143. Age comparison supports targeted campaigns rather than treating the customer base as homogeneous.

Power BI Dashboard

KPI Cards

3.9K customers · \$59.76 avg purchase · 3.75 avg rating

Category Visuals

Clothing leads revenue (~\$100K), followed by Accessories, Footwear, Outerwear

Age Group Visuals

Young Adult and Middle-aged drive the highest revenue and sales

Interactive Slicers

Subscription Status, Gender, Category, and Shipping Type filters

Business Recommendations



Boost Subscriptions

Promote exclusive benefits and clear value propositions to grow the subscriber base.



Strengthen Loyalty Programs

Reward repeat buyers and create pathways to move Returning customers into the Loyal segment.



Review Discount Policy

Use product-level discount dependence data to balance sales stimulation against margin protection.



Target by Age & Fulfillment

Use higher-revenue age groups and shipping-behavior patterns to make marketing more focused.



Project Conclusion

This project demonstrates a complete end-to-end analytics workflow: raw transactional data is prepared in Python, business questions are answered with SQL Server, and results are communicated through Power BI.

- The strongest value of this project is not any single chart – it is the connection between data preparation, reproducible SQL analysis, and decision-oriented visualization.

Python

Cleaning & feature engineering

SQL Server

Business aggregation

Power BI

Interactive dashboard